

SUPRIYA MANDAL

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Profile Summary

Backend and data Engineer with over 4 years of experience building high throughput, **event-driven systems** and streaming data pipelines in production **SaaS** environments. Focus on building simple cloud-native, reliable, observable systems at scale.

Education

Indian Institute of Science (IISc)

M.Tech in Computational and Data Sciences

Bangalore

2021 – 2023

Sathyabama Institute of Science and Technology

B.Tech in Information Technology

Chennai

2016 – 2020

Experience

Senior Software Engineer (ML)

May 2026 – Present

Chargebee

Bangalore

- Owned and productionized a **Churn prediction** ML platform live for **200+ merchants**, from feature pipeline through model serving.
- Contributed in building batch data pipelines with **PySpark** and **Airflow DAGs**, processing merchant billing/usage events into ML-ready features

Software Engineer (Platform)

Aug 2023 – April 2026

Chargebee

Bangalore

- Owned the central **Queue Management System** (AWS SQS/SNS), managing end-to-end asynchronous workflows across multiple product teams; became top maintainer of the system.
- Built core components of a **Kafka** based event-driven platform handling millions of webhook delivery at scale; improved the webhook delivery success rate by implementing endpoint level **rate limiting** using **Redis**.
- Developed **AWS MSK**, **ECS** baseline infrastructure with producer/consumer implementations, ensuring reliability and observability using Datadog and Splunk.
- Reduced deployment time from **2-3 days** to under **1 hour** by designing CI/CD pipelines to automate queue provisioning and management.
- Automated infrastructure provisioning using **Terraform/Terragrunt**.
- Partnered with internal teams [CPE, Infra, Subscription etc] to onboard asynchronous flows, help them to design, debugging, and testing their integrations using our messaging framework.
- Participated in production incident response, root cause analysis, and implemented preventive measures for recurring patterns.

Computer Vision Researcher

March 2022 – July 2023

Visual Computing Lab @ IISc

Bangalore

- Developed **unsupervised Deep Learning** models for video anomaly detection and localization.
- Built few-shot **face recognition** system using Siamese Neural Networks.
- Fine-tuned CNN architectures across multiple datasets for improved model performance.

Projects

DLL Based Video Anomaly Detection [Link] | *Python, Tensorflow, Pytorch, OpenCV*

Dec 2022 – June 2023

- Developed video anomaly detection and localization model using **CNN-LSTM** and **3D Convolution autoencoder**.
- Did experimentations with **Swin Transformer** to reformulate video classification as image classification.
- Trained and tested the model in an unsupervised manner using UCSD pedestrian dataset.

Few Shot Face Recognition [Link] | *Python, Tensorflow, Keras, Scikit Learn*

April 2022

- Built face recognition system using **Siamese Neural Networks** to identify faces with minimal training samples.
- Used very limited data for each category to train the model

Technical Skills

Languages and Frameworks: Java, Go, Python, SQL, Spring Boot

Distributed systems & streaming: Apache Kafka, AWS (SQS/SNS, MSK, ECS, EC2, DynamoDB, S3, EMR, Batch), PySpark, Airflow

Cloud & Infrastructure: Docker, Terraform, Git, GitHub Actions, AWS CodeBuild

Monitoring and Observability: Datadog, Splunk, AWS CloudWatch

AI tools: Cursor, Claude code, Claude